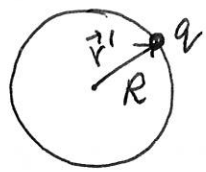
P1.

<u>Electric dipole:</u> $v = \omega R$



$$\vec{r}'(t) = R\left(\cos(\omega t+\alpha)\,\hat{x} + \sin(\omega t+\alpha)\,\hat{y}\right) \qquad \alpha = \text{initial phase}$$

$$\dot{\vec{r}}' = R\left(-\omega \sin(\omega t+\alpha)\,\hat{x} + \omega\cos(\omega t+\alpha)\,\hat{y}\right)$$

$$\ddot{\vec{r}}' = -\omega^2 \vec{r}' \quad , \quad \vec{d} = q\vec{r}'$$

$$\vec{B} = \frac{1}{c}\frac{\mu_0}{4\pi r}\, q\,\ddot{\vec{r}}' \times \hat{n} = -\frac{\mu_0 q \omega^2}{4\pi c r}\,\vec{r}' \times \hat{n}\,\Big|_{t'=t-\frac{r}{c}}$$

$$= -\frac{\mu_0 q \omega^2}{4\pi c r}\left\{\frac{\sin\left[\omega(t-\frac{r}{c})+\alpha\right] z\,\hat{x} - \cos\left[\omega(t-\frac{r}{c})+\alpha\right] z\,\hat{y}}{r}\right.$$

$$\left. + \frac{\cos\left[\omega(t-\frac{r}{c})+\alpha\right] y - \sin\left[\omega(t-\frac{r}{c})+\alpha\right] x}{r}\,\hat{z}\right\}$$

where we used $\hat{n} = \frac{\vec{r}}{r} = \frac{x}{r}\hat{x} + \frac{y}{r}\hat{y} + \frac{z}{r}\hat{z}$

$$\vec{E} = c\,\vec{B} \times \hat{n}$$

<u>Magnetic dipole moment</u> contributes nothing.

$$\vec{m} = \frac{q}{2}\,\vec{r}'(t) \times \vec{v}(t)$$

$$\dot{\vec{m}} = \frac{q}{2}\left(\vec{v} \times \vec{v} + \vec{r}' \times \vec{a}\right) = 0.$$

but $\vec{a}$ = centripetal acceleration along radial direction $\vec{r}'$

$$\frac{dI}{d\Omega} = \frac{c}{\mu_0}\vec{B}^2 r^2 = \frac{\mu_0}{c}\left(\frac{q\omega^2}{4\pi}\right)^2\left\{\frac{z^2}{r^2}\left(\sin^2[\;] + \cos^2[\;]\right) + \right.$$

$$\left. + \left(\frac{-x\sin[\;] + y\cos[\;]}{r}\right)^2\right\} \qquad [\;] \equiv \omega\left(t-\frac{r}{c}\right)+\alpha$$

But $\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases}$

$$\Rightarrow \frac{x \sin[\omega t' + \alpha] - y \cos[\omega t' + \alpha]}{r} = \sin\theta \left(\cos\varphi \sin[\omega t' + \alpha] - \sin\varphi \cos[\omega t' + \alpha] \right)$$

$$= \sin\theta \sin[\omega t' + \alpha - \varphi]$$

$$\frac{dI}{d\Omega} = \frac{\mu_0 q^2 \omega^4}{16\pi^2 c} \left\{ \overset{\cos^2\theta}{\cancel{1}} + \underbrace{\sin^2}_{\sin^2\theta}\left[\omega\left(t - \frac{r}{c}\right) + \alpha - \varphi\right] \right\}$$

Average over the period of rotation: $T = \frac{2\pi}{\omega}$, $\xi = \omega t$

$$\frac{1}{T}\int_0^T dt \, \sin^2\left[\omega\left(t - \frac{r}{c}\right) + \alpha - \varphi\right] = \int_0^{2\pi} \frac{d\xi}{2\pi} \sin^2\left[\xi - \frac{\omega r}{c} + \alpha - \varphi\right] = \frac{1}{2}.$$

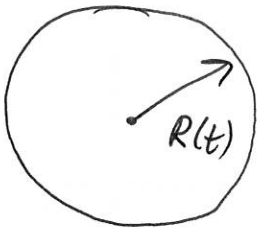
$$\Rightarrow \left\langle \frac{dI}{d\Omega} \right\rangle = \frac{\mu_0 q^2 \omega^4}{16\pi^2 c} \left\{ \cos^2\theta + \frac{1}{2}\sin^2\theta \right\} = \frac{\mu_0 q^2 \omega^4}{16\pi^2 c} \left(1 - \frac{1}{2}\sin^2\theta \right)$$

$$P = \int \frac{dI}{d\Omega} d\Omega = \frac{\mu_0 q^2 \omega^4}{16\pi^2 c} \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \left(1 - \frac{1}{2}\sin^2\theta \right) =$$

$$= \frac{\mu_0 q^2 \omega^4}{4\pi c} \int_0^1 d(\cos\theta) \underbrace{\left(1 - \frac{1}{2}(1 - \cos^2\theta) \right)}_{\frac{1}{2}(1 + \cos^2\theta)} =$$

$$= \frac{\mu_0 q^2 \omega^4}{4\pi c} \frac{1}{2}\left(1 + \frac{1}{3}\right) = \frac{\mu_0 q^2 \omega^4}{6\pi c}.$$

P2.



due to $\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$

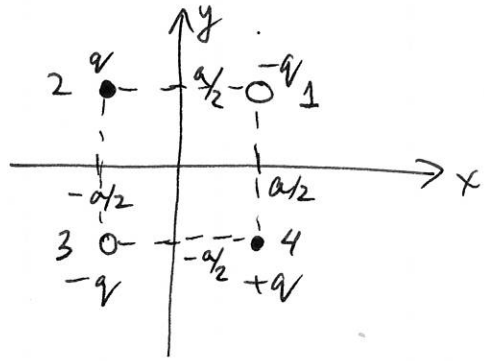
Electric field $\vec{E}$ of oscillating sphere is <u>strictly</u> radial:

$$\vec{E}(\vec{r},t) = \frac{Q}{4\pi\varepsilon_0 r^2}\,\hat{r}$$

$$\Rightarrow \vec{B} = \frac{1}{c}\,\hat{n}\times\vec{E} \sim \hat{r}\times\hat{r} = 0$$

$\Rightarrow$ no radiation!

P3 a)



$$a(t) = a_0 \cos(\omega t)$$

$$\vec{x}_j^{\,2} = \left(\frac{a}{2}\right)^2 + \left(\frac{a}{2}\right)^2 = \frac{a^2}{2}$$

Quadrupole moment $Q_{\alpha\beta} = \sum_{j=1}^{4} q_j \left(3\, x_{j\alpha}\, x_{j\beta} - \delta_{\alpha\beta}\, x_j^2\right) =$

$$= -\delta_{\alpha\beta} \underbrace{\sum_{j=1}^{4} q_j \left(\frac{a^2}{2}\right)}_{=0} + 3q\left(x_{2\alpha}\, x_{2\beta} + x_{4\alpha}\, x_{4\beta} - x_{1\alpha}\, x_{1\beta} - x_{3\alpha}\, x_{3\beta}\right)$$

$$Q_{xx} = 0 = Q_{yy}\,, \quad Q_{zz} = 0$$

$$Q_{xy} = 3q\left(-\frac{a^2}{4}\cdot 4\right) = -3qa^2. = Q_{yx}$$

$$\Rightarrow \hat{Q} = (-3qa^2)\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\vec{A} = \frac{\mu_0}{4\pi r}\,\frac{1}{6c}\,\ddot{\vec{Q}}\,, \quad \text{vector } Q_i = Q_{ij}\,\hat{n}_j$$

$$Q_x = Q_{xx}\,\hat{n}_x + Q_{xy}\,\hat{n}_y + Q_{xz}\,\hat{n}_z = Q_{xy}\, n_y$$

$$Q_y = Q_{xy}\, n_x$$

$$\Rightarrow \vec{Q} = Q_{xy}\,(n_y, n_x, 0) = (-3qa^2)\,(n_y, n_x, 0)$$

$$\vec{B} = \frac{\mu_0}{24\pi c^2 r}\,\dddot{\vec{Q}} \times \hat{n} = \frac{\mu_0}{24\pi c^2 r}\,\frac{d^3(-3qa^2(t))}{dt^2}\,(n_y, n_x, 0) \times (n_x, n_y, n_z)$$

$$= \frac{-\mu_0 q}{8\pi c^2 r}\left(\frac{d^3 a^2}{dt^3}\right)\left\{n_x n_z\,\hat{x} - n_y n_z\,\hat{y} + (n_y^2 - n_x^2)\,\hat{z}\right\}$$

$$\frac{d^3 a^2}{dt^3} = \frac{d^2}{dt^2}\left(2 a \dot{a}\right) = 2 \frac{d}{dt}\left(\dot{a}\dot{a} + a\ddot{a}\right) =$$

$$= 2\left(2\dot{a}\ddot{a} + \dot{a}\ddot{a} + a\dddot{a}\right) = 2\left(3\dot{a}\ddot{a} + a\dddot{a}\right) =$$

$$= 2 a_0^2\left[-3\omega \sin(\omega t)\left(-\omega^2 \cos(\omega t)\right) + \cos(\omega t)\left(\omega^3 \sin(\omega t)\right)\right]$$

$$= 2 a_0^2 \omega^3\left[3\sin(\omega t)\cos(\omega t) + \cos(\omega t)\sin(\omega t)\right] =$$

$$= 8 a_0^2 \omega^3 \sin(\omega t)\cos(\omega t)$$

$$\vec{B}(r,t) = \frac{-\mu_0 q a_0^2 \omega^3}{\pi c^2 r} \cos\left[\omega\left(t-\frac{r}{c}\right)\right]\sin\left[\omega\left(t-\frac{r}{c}\right)\right] \cdot$$

$$\cdot \frac{xz\,\hat{x} - yz\,\hat{y} + (y^2-x^2)\,\hat{z}}{r^2}$$

$$\vec{E} = c\,\vec{B}\times\hat{n}\ , \quad \hat{n} = \frac{\vec{r}}{r} = \frac{x}{r}\hat{x} + \frac{y}{r}\hat{y} + \frac{z}{r}\hat{z}.$$

$$\frac{dI}{d\Omega} = \frac{c}{\mu_0}\vec{B}^2 r^2$$

Average over period $T = \frac{2\pi}{\omega}$: $\frac{1}{T}\int_0^T dt\, \cos^2\left[\omega\left(t-\frac{r}{c}\right)\right]\sin^2\left[\omega\left(t-\frac{r}{c}\right)\right] =$

$\xi = \omega t$ $\quad = \frac{1}{4}\int_0^{2\pi}\frac{d\xi}{2\pi}\sin^2\left(2\xi - \frac{\omega r}{c}\right) = \frac{1}{4}\cdot\frac{1}{2} = \frac{1}{8}.$

$$\frac{dI}{d\Omega} = \frac{\mu_0 q^2 a_0^4 \omega^6}{8\pi c^3}\,\frac{x^2z^2 + y^2z^2 + (x^2-y^2)^2}{r^4} =$$

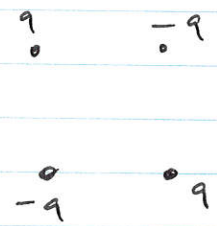
$$= \frac{\mu_0 q^2 a_0^4 \omega^6}{8\pi c^3}\left[\cos^2\theta\sin^2\theta + \sin^4\theta\,(\cos 2\varphi)^2\right].$$

$$P = \int \frac{dI}{d\Omega} \sin\theta \, d\theta \, d\varphi = \frac{\mu_0 q^2 a_0^4 \omega^6}{8\pi c^3} \cdot \pi \cdot$$

$$\cdot \underbrace{\int_0^{\pi} d\theta \sin\theta \left[2\cos^2\theta \sin^2\theta + \sin^4\theta \right]}_{8/5}$$

$$\Rightarrow \quad P = \frac{\mu_0 q^2 a_0^4 \omega^6}{5c^3}$$

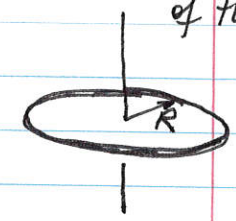
total power radiated by quadrupole made of 4 point charges



P3b (b) Quadrupole moment of charge distribution

$$Q_{ij} = \int \rho(r')\left(3 r_i' r_j' - \delta_{ij}\, \vec{r}'^2\right) d^3\vec{r}'$$ eq. 10.64 of the Brau's book

Charged ring $\quad \rho = \frac{q}{2\pi R}\,\delta(\rho - R)\,\delta(z)$



Due to cylindrical symmetry of the distribution

$$Q_{xx} = Q_{yy} \quad \text{while} \quad Q_{xy} = Q_{yx} = 0 \left[\text{because} \int_0^{2\pi} d\varphi \cos\varphi \sin\varphi = 0\right]$$

Total charge $= \frac{q}{2\pi R} \underbrace{\int dz \int_0^{2\pi} d\varphi \int_0^{\infty} d\rho\, \rho}_{d^3\vec{r}'}\, \delta(z)\,\delta(\rho - R) = q$

$$Q_{xx} = \frac{q}{2\pi R} \int dz \int_0^{2\pi} d\varphi \int d\rho\, \rho \left(3x^2 - (x^2+y^2+z^2)\right) \delta(z)\,\delta(\rho - R)$$

$$= \frac{q}{2\pi R} \int_0^{2\pi} d\varphi \int d\rho\, \rho \left(3\rho^2 \cos^2\varphi - \rho^2\right) \delta(\rho - R)$$

$$= \frac{q}{2\pi R} \int d\rho\, \rho\, \delta(\rho - R) \left(3\rho^2 \cdot \pi - \rho^2 \cdot 2\pi\right)$$

$$= \frac{q}{2\pi R} \int d\rho\, \rho^3 \pi\, \delta(\rho - R) = \frac{q\,\pi R^3}{2\pi R} = \frac{1}{2} q R^2.$$

$$Q_{zz} = -\frac{q}{2\pi R} \int dz\, \delta(z) \int_0^{2\pi} d\varphi \int d\rho\, \rho\, \delta(\rho - R) \left[\rho^2 + z^2\right] =$$

$$= -\frac{q}{2\pi R} \cdot 2\pi \cdot R^3 = -qR^2 = -2Q_{xx}.$$

$$\hat{Q} = Q_{xx} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \quad \text{(traceless)}$$

$$\vec{B} = \frac{\mu_0}{24\pi c^2 r} \dddot{\vec{Q}} \times \hat{n}$$

$$\vec{Q} = Q_{xx}\left(n_x, n_y, -2n_z\right)$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ n_x & n_y & -2n_z \\ n_x & n_y & n_z \end{vmatrix} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & -3n_z \\ n_x & n_y & n_z \end{vmatrix} = 3 n_y n_z \hat{x} - 3 n_x n_z \hat{y}.$$

$$\Rightarrow \quad \vec{B} = \frac{\mu_0}{8\pi c^2 r} \dddot{Q}_{xx}\left(n_y n_z \hat{x} - n_x n_z \hat{y}\right)$$

$\vec{E} = c\,\vec{B} \times \hat{n}$ is along $\hat{z}$ axis.

$$\frac{dI}{d\Omega} = \frac{c}{\mu_0} \vec{B}^2 r^2 = \frac{c}{\mu_0}\left(\frac{\mu_0}{8\pi c^2}\right)^2 \left(\dddot{Q}_{xx}\right)^2 \underbrace{n_z^2\left(n_y^2 + n_x^2\right)}_{\cos^2\theta \sin^2\theta}$$

$$\dddot{Q}_{xx} = \frac{1}{2} q \left(\frac{d^3}{dt^3} R^2\right) = q \frac{d^2}{dt^2} R\dot{R} = q \frac{d}{dt}\left(\dot{R}^2 + R\ddot{R}\right) =$$

$$= q\left(2\dot{R}\ddot{R} + \dot{R}\ddot{R} + R\dddot{R}\right)$$

$$= q\left(3\dot{R}\ddot{R} + R\dddot{R}\right)$$

$$\frac{dI}{d\Omega} = \frac{\mu_0 q^2}{64\pi^2 c^3}\left(3\dot{R}\ddot{R} + R\dddot{R}\right)^2 \sin^2\theta \cos^2\theta$$

$$\int d\Omega \sin^2\theta \cos^2\theta = 2\pi \cdot \frac{4}{15}$$

$$\Rightarrow P = \int \frac{dI}{d\Omega} d\Omega = \frac{\mu_0 q^2}{120\pi c^3}\left(3\dot{R}\ddot{R} + R\dddot{R}\right)^2.$$